Vipul Sapra

Charlotte, North Carolina | 513-693-6897 | vpsapra5@gmail.com | linkedin.com/in/vipul-sapra-ab30a09a

EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Supervised and Unsupervised Learning, Agentic AI Workflows, LLM Prompt Orchestration, Autonomous Anomaly Detection, Predictive Analytics, Advanced Time-Series Modeling (SARIMA), Data Lakehouse Architecture, Dimensional Modeling, Automated Data Reconciliation, MLOps, Statistical Analysis

Programming Languages & Tools: Python, PySpark, XGBoost, Scikit-Learn, SARIMA, AWS (S3, Redshift, Lambda, Athena), Azure (ADF, Databricks, Synapse, Azure SQL), Snowflake, Tableau, Power BI (DAX), KIRO IDE, Collibra, Apache Spark

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data Scientist - Analytics and Insights Engineer III — **May 2022 – Present**

- **Architected autonomous agentic AI workflows** utilizing KIRO IDE, prompt orchestration, and LLMs to automate complex schema discovery, data profiling, and code migration, reducing manual delivery turnaround by 40-50%.
- **Engineered autonomous machine learning anomaly detection** frameworks identifying schema variations, distribution drift, and ETL discrepancies with 95% precision, safeguarding downstream BI and pipeline integrity.
- **Developed predictive machine learning models** using XGBoost, Random Forests, and Scikit-Learn to forecast customer behavior, delivering actionable quantitative insights that improved service reliability by 20%.
- **Built advanced time-series forecasting models** via SARIMA to predict truck rolls and field support demand, optimizing workforce capacity allocation and reducing operational response times by 20%.
- **Engineered automated testing and evaluation frameworks** for enterprise conversational AI platforms (SpectrumIQ), cutting manual QA turnaround by 50% and accelerating production release cycles for intelligent virtual assistants.

Data Scientist & Cloud Data Platform Engineer — **May 2022 – Present**

- **Architected distributed pipelines and dimensional models** (Star/Snowflake schemas, SCD Type 1/2) using Azure (ADF, Databricks, Synapse) and AWS (S3, Redshift) to process hundreds of terabytes with 99.9% data availability.
- **Engineered high-throughput PySpark and Apache Spark** transformations on Snowflake and cloud lakehouses to unify disparate client and transactional datasets for scalable analytics.
- **Built automated source-to-target reconciliation firewalls** and Collibra data lineage solutions, reducing operational turnaround by over 80% and ensuring audit-ready data governance.

Business Intelligence & Analytics Solutions Engineer — **May 2022 – Present**

- **Designed executive Key Operating Metric (KOM) dashboards** and semantic models in Power BI (DAX) and Tableau, translating complex transactional telemetry into actionable corporate strategy.
- **Translated complex quantitative concepts** into clear executive-level insights, enabling non-technical stakeholders to make data-driven decisions on operational performance.
- **Conducted rigorous statistical analysis** and data profiling on large-scale datasets, ensuring rigorous data quality and adherence to enterprise reporting standards.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.